

Quiz 15% (n-1)

Assignment 5%

Mid 25%

Final 30%

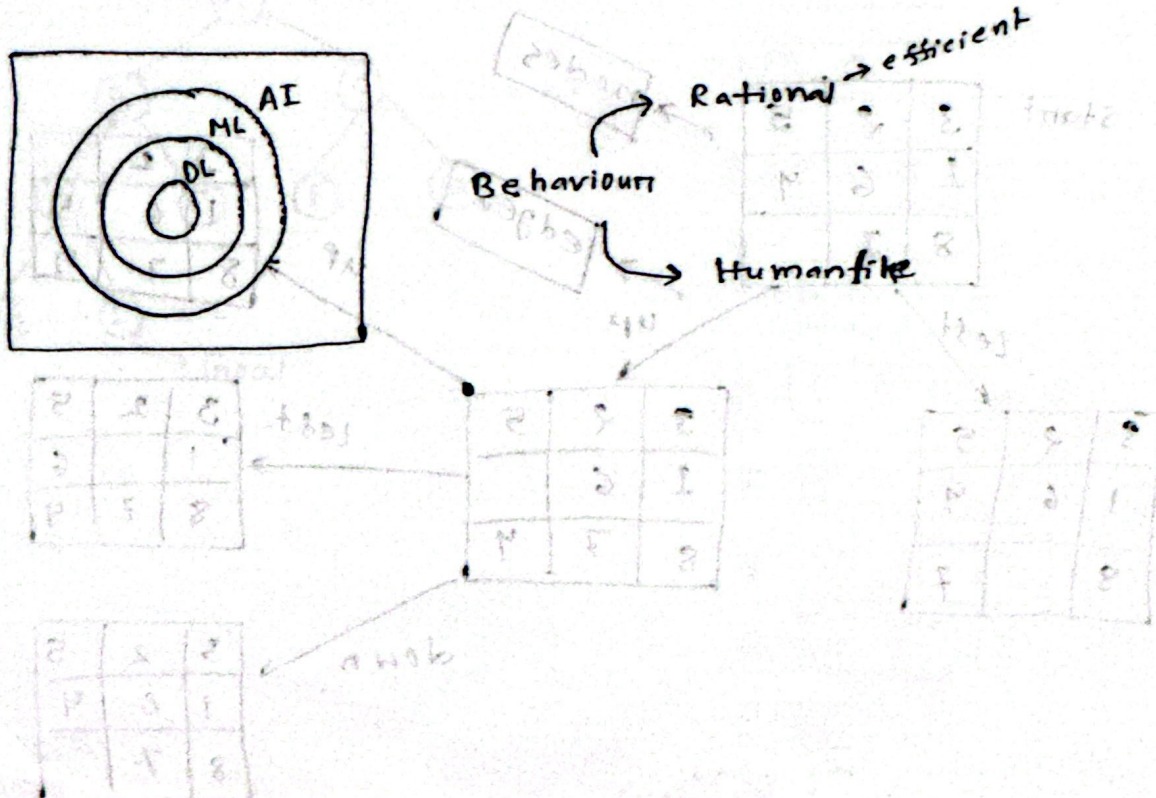
Lab 25%

Artificial Intelligence

- i) Think
- ii) Problem-solve
- iii) communicate
- iv) Learn [store knowledge]
- v) Emotion

AI term propose - 1956

1950s → Turing Test



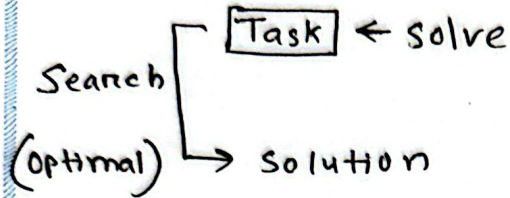
TOPIC NAME : Searching

DAY :

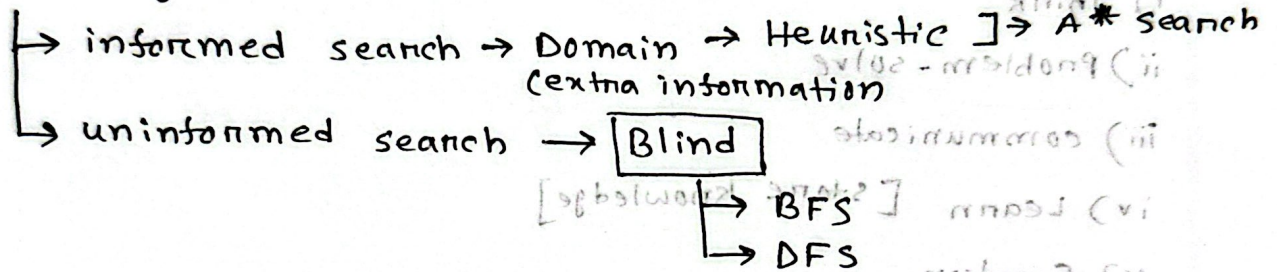
TIME :

DATE 21 / 6 / 25

Agent

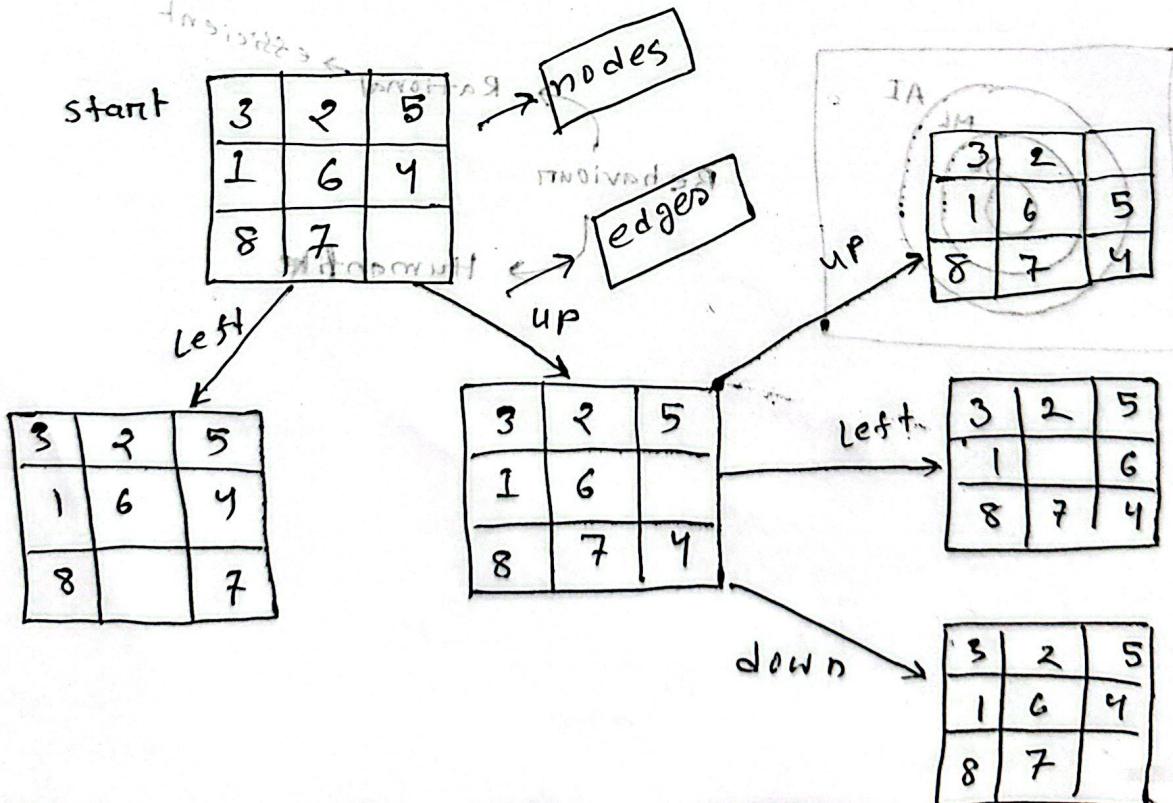


Searching



Uninformed searching

example : 8 puzzle game



TOPIC NAME : _____

DAY : _____

TIME : _____

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3	2	5
1	6	4
8		7

left

right

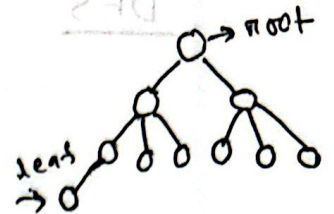
up

3	2	5
1	6	4
	8	7

3	2	5
1	6	4
8	7	

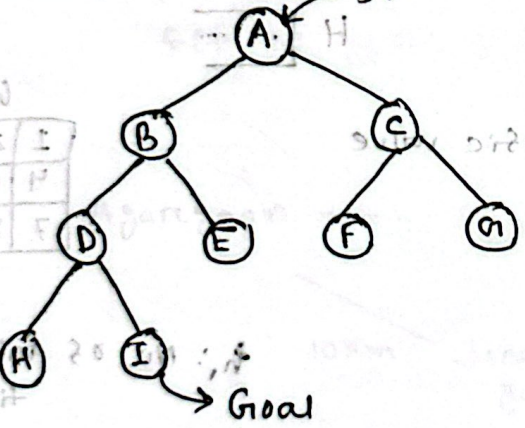
3	2	5
1		4
8	6	7

state space tree



BFS → level wise

start



Fringe 'Queue'

A

A B C

B C D E

C D E F G

D E F G H I

E F G H I

F G H I

G H I

H I

Goal

3	2	5
1	6	4
8		7

3	2	5
1	6	4
8	7	

3	2	5
1	6	4
8	7	

H I

Goal

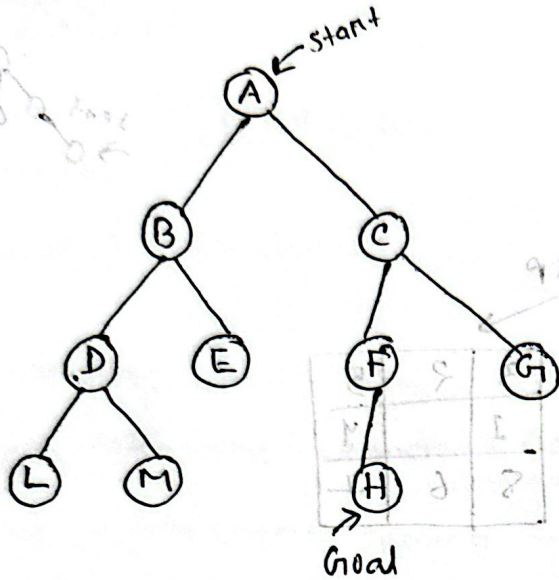
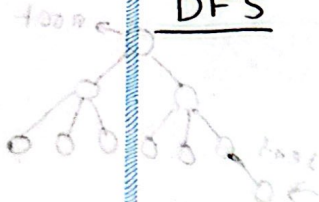
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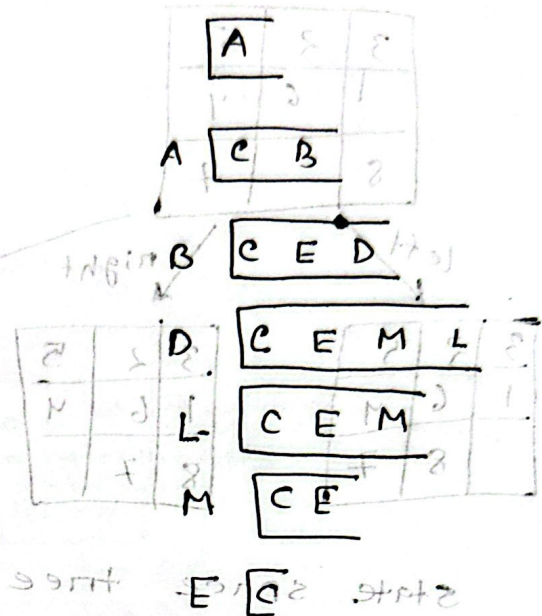
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DFS



Fringe $\rightarrow$ stack

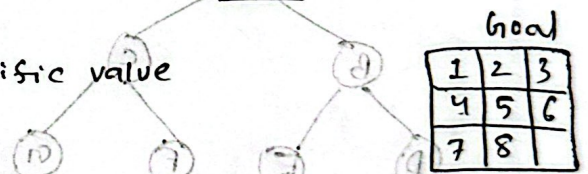


Optimal solution $\rightarrow$ H

Informed searching

$\rightarrow$ Heuristic value

8 puzzle problem



1	2	3
5		8
4	6	7

left

up

right

down

1	2	3
	5	8
4	6	7

$$h_1 = 4$$

$$h_2 = 2 + 1 + 2 + 2$$

1		3
5	2	8
4	6	7

$$h_1 = 6$$

$$h_2 = 9$$

1	2	3
5	8	
4	6	7

$$h_1 = 5$$

$$h_2 = 8$$

1	2	3
5	6	8
4		7

$$h_1 = 1 + 1 + 1 + 1 + 1 = 5$$

$h_2 =$ Manhattan distance

$$h_2 = 1 + 2 + 1 + 2 + 2 = 8$$

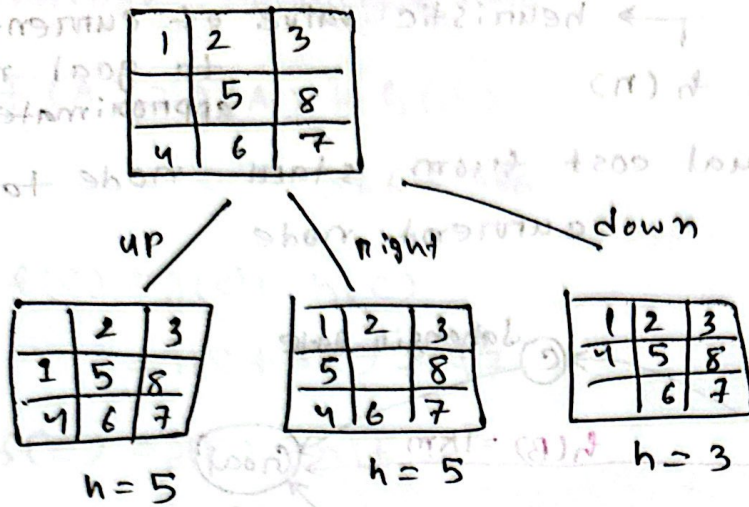
h_1 : No. of misplaced tile

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /



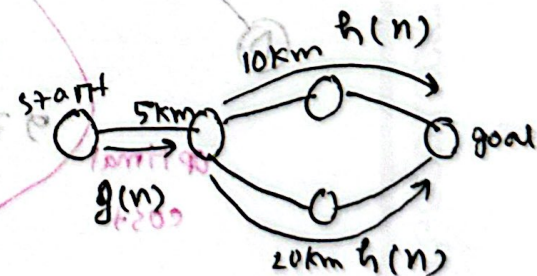
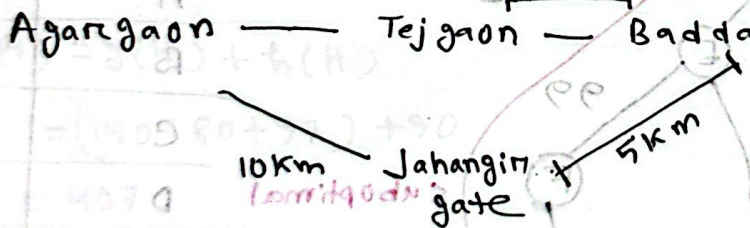
A* search

Heuristic value + Actual cost

$h(n) \rightarrow$ heuristic value

$g(n) \rightarrow$ actual cost from start node to current node

$$f(n) = g(n) + h(n)$$



$g(n) + h(n)$ is the distance of path used

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DAY : _____

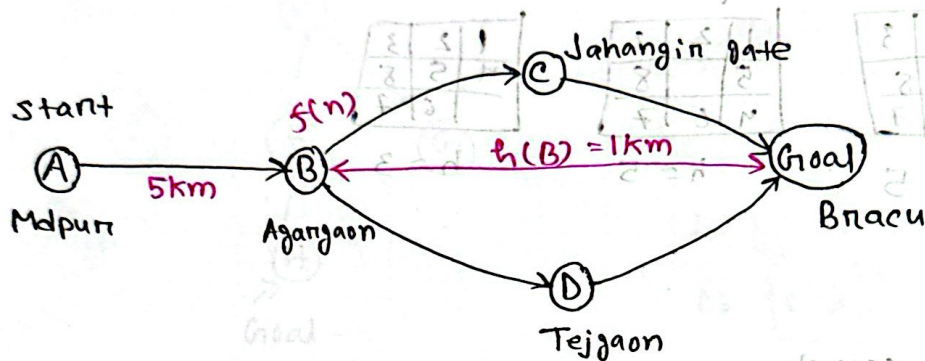
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DATE : 26 / 6 / 25

A* search

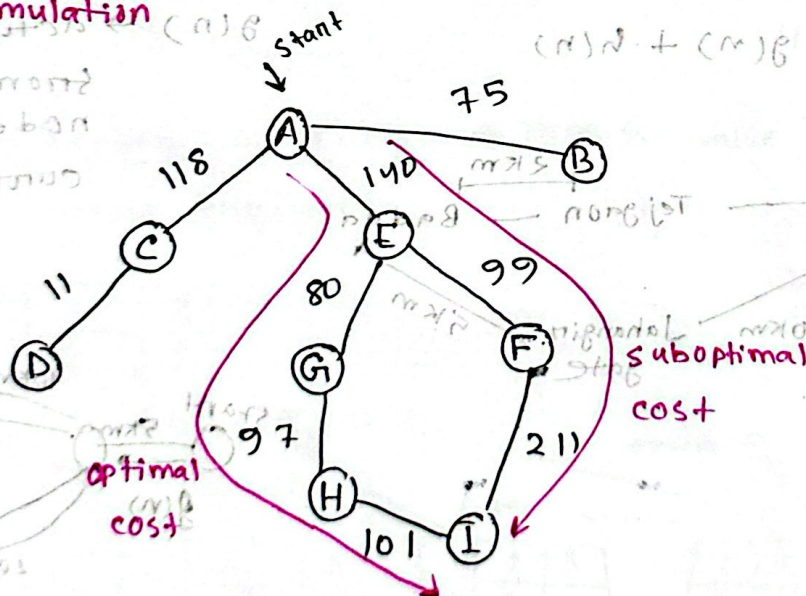
$$f(n) = g(n) + h(n)$$

→ heuristic value of current node to goal node / approximate value
 ↳ actual cost from start node to current node



$$f(B) = 5 \text{ km} + h(B) = 5 \text{ km} + 1 \text{ km}$$

Simulation



Heuristic values

Nodes	$h(n)$
A	366
B	374
C	329
D	244
E	253
F	178
G	193
H	90
I	0

$h(n) = 0$ when reached goal

Simulate using A* search.

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Fringe : Priority Queue

Tree Search (A* Search)

$$f(A) = g(A) + h(A)$$

$$= 0 + 366 = 366$$

$$f(C) = g(C) + h(C)$$

$$= 118 + 329 = 447$$

$$f(E) = g(E) + h(E)$$

$$= 140 + 253 = 393$$

$$f(B) = g(B) + h(B)$$

$$= 75 + 374 = 449$$

$$f(G) = g(G) + h(G)$$

$$= (140 + 80) + 193 = 413$$

$$f(F) = g(F) + h(F)$$

$$= (140 + 99) + 178$$

$$= 417$$

$$f(H) = g(H) + h(H)$$

$$= (140 + 80 + 97) + 90$$

$$= 407$$

$$f(I) = g(I) + h(I)$$

$$= (140 + 80 + 97 + 101) + 0$$

$$= 418$$

$$f(J) = g(J) + h(J)$$

$$= (140 + 99 + 211) + 0$$

$$= 450$$

A. 366

A 366 C 447 E 393 B 449

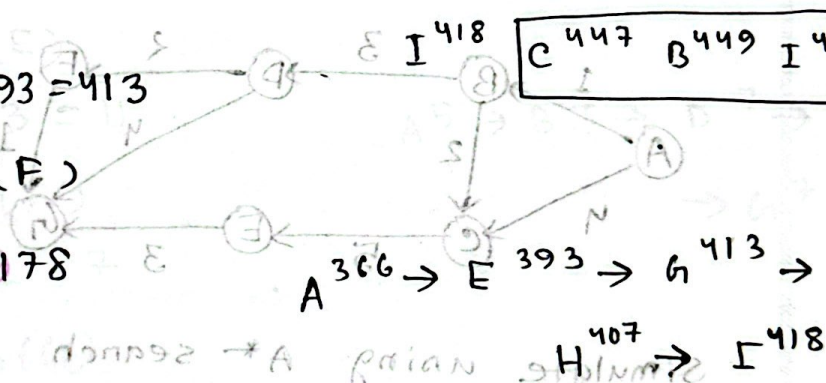
E 393 C 447 B 449 G 413 F

G 413 C 447 B 449 F 417 H 407

H 407 C 447 B 449 F 417 I 418

F 417 C 447 B 449 I 418 I 450

I 418 C 447 B 449 I 450



TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Graph search (A* search)

↳ visited list

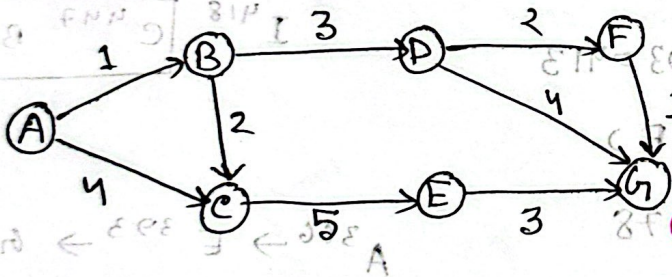
একটা node দুইবার থাকলে
একটা খসত delete করতে,
অন্যটা remove করে দিতে

→ একবার বের করে থাকা
আর কখনোই enqueue
করতে না।

Tree search (A* search)

একটা node expand করলেও
node ও বের করে দিতে।

Q2:



Simulate using A* search

(tree search)

node	$h(n)$
A	5
B	6
C	4
D	3
E	3
F	1
G	0

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

$$F(A) = g(A) + h(A)$$

$$= 0 + 5 = 5$$

A^5

$$F(B) = g(B) + h(B)$$

$$= 1 + 6 = 7$$

A^5

$B^7 C^8$

$$F(C) = g(C) + h(C)$$

$$= 4 + 4 = 8$$

B^7

$C^8 C^7 D^7$

$$F(C) = g(C) + h(C)$$

$$= (1+2) + 4 = 7$$

C^7

$C^8 D^7 E''$

$$F(D) = g(D) + h(D)$$

$$= (1+3) + 3 = 7$$

D^7

$C^8 E'' F^7 G^8$

$$F(E) = g(E) + h(E)$$

$$= (1+2+5) + 3 = 11$$

F^7

$C^8 E'' G^8 G^7$

$$F(F) = g(F) + h(F)$$

$$= (1+3+2) + 1 = 7$$

G^7

$C^8 E'' G^8$

$$F(G) = g(G) + h(G)$$

$$= (1+3+4) + 0 = 8$$

$$F(G) = g(G) + h(G)$$

$$= (1+3+2+1) + 0 = 7$$



$A^5 \rightarrow B^7 \rightarrow D^7 \rightarrow F^7 \rightarrow G^7$

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

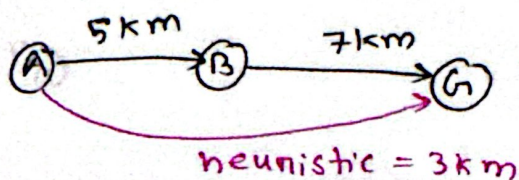
Admissibility:

$$h(n) \leq h^*(n)$$

~~n ના પાઠ Goal ના અંતર અપ્રોક્સિ~~

$h(n)$ = current node ના Goal ના અંતર
 $\therefore$ heuristic value

$h^*(n)$ = current node ના Goal ના અંતર -
 actual cost



$$h(n) \leq h^*(n)$$

$$3\text{km} \leq 12\text{km}$$

So admissible

↳ optimal solution ના

optimal soln ના શરૂ overestimate કર્યા થાય ના,

so heuristic value $\leq h^*(n)$ (રજા શરૂ

↳ actual cost

TOPIC NAME : _____

DAY : _____

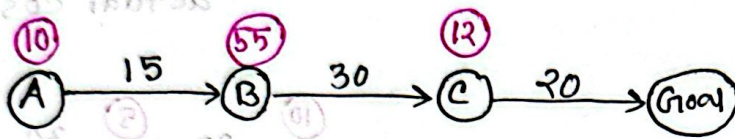
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Heuristic Admissibility

For every node n

$$h(n) \leq h^*(n)$$

$h^*(n)$ = actual cost from current node to goal node

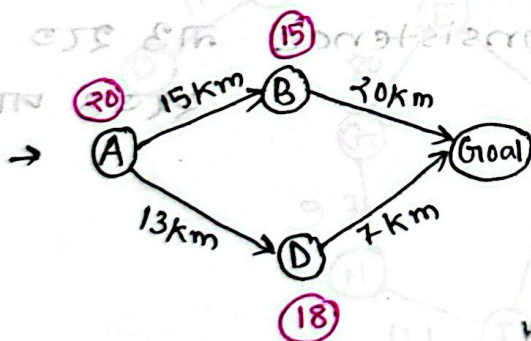


$$h(B) \leq h^*(B)$$

$$55 \leq 50$$

overestimated heuristic value $55 \geq 50$

पूरा सिस्टम पर सब node heuristic admissible रन system पर admissible.



$$h(A) \leq h^*(A)$$

$$20 \leq 35$$

$$20 \leq 20$$

$$h(D) \leq h^*(D)$$

$18 \leq 7$ X not admissible

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Consistency: $\rightarrow$ efficiently optimal solution

$$h(n) \leq c(n, n') + h(n')$$

$$c(n, n')$$

(current node, child node)
actual cost

$$h(A) = 10$$

$$c(A, B) = 20$$

$$h(B) = 5$$

$$10 \leq 20 + 5$$

$$10 \leq 25$$

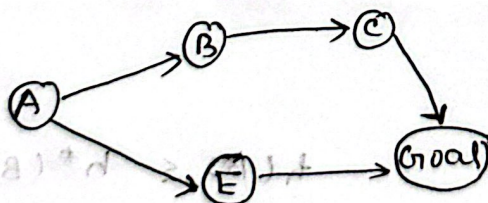
Admissible $\rightarrow$ optimal soln

consistency: efficiently optimal soln

consistence হল admissible হবে।

কিন্তু admissible হল consistency নাও হতে পারে,

হতেও পারে।

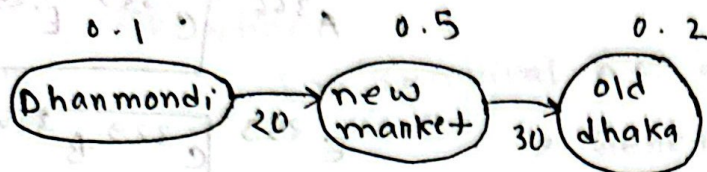


$$h(A) \leq c(A, B) + h(B)$$

$$h(A) \leq c(A, E) + h(E)$$

TOPIC NAME: _____

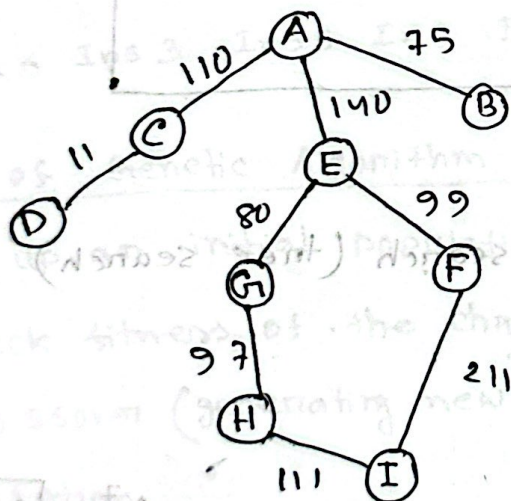
Heuristic Dominance :



h_1	h_2
20	1.1
50	2.1
60	3.1
90	3.5
100	2.8
120	0.5

compare difficulty
we'll take h_1 in this case.

Greedy Best First Search (GBFS) : → Informed Search
not optimal



A	366
B	374
C	329
D	244
E	253
F	178
G	193
H	90
I	0

Tree GBFS → not optimal, because this will stuck in infinite loop (not complete)

Graph search prevents loop.

Graph → finite (complete), infinite (incomplete)

TOPIC NAME : _____

DAY : _____

TIME : _____

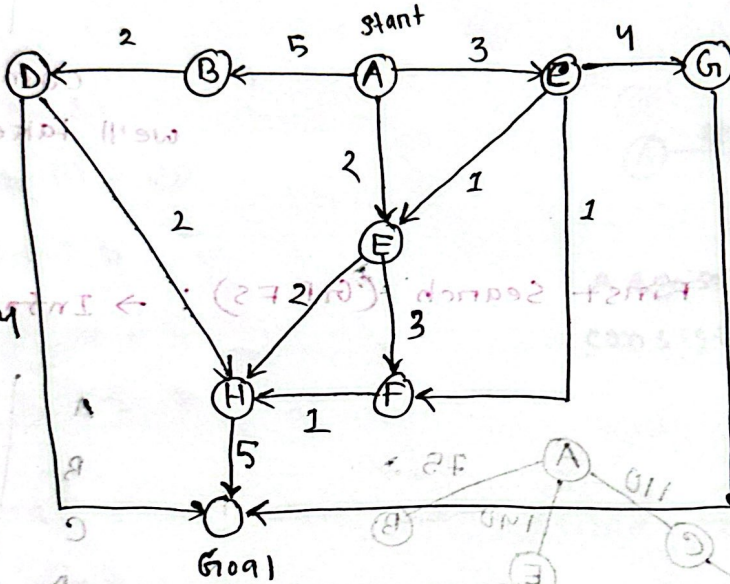
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$$f(n) = h(n)$$

A ³⁶⁶

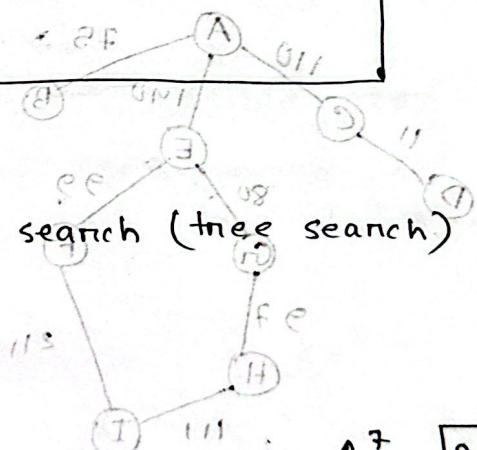
C ³²⁹ E ²⁵³ B ³⁷⁴

C ³²⁹ B ³⁷⁴ G ¹⁹³ F ¹⁷⁸



n	h(n)
A	7
B	5
C	6
D	3
E	4
F	4
G	2
H	2
Goal	0

Simulate using A* search (tree search)



A ⁷

B ¹⁰ C ⁹ E ⁶

E ⁶ B ¹⁰ C ⁹ F ⁹ H ⁶

H ⁶ B ¹⁰ C ⁹ F ⁹ Goal ⁰

Goal ⁰ B ¹⁰ C ⁹ F ⁹

Genetic Algorithm

Evolution

↳ Survival of the fittest (stronger survive longer)

↳ Genetic Algo based on this theory

Insect 1

Insect 2

Insect 3

~~Insect 4~~

Ins 1

Ins 2

Ins 3

Ins 5

Ins 6

Ins 7

Ins 8

Ins 1

Ins 2

Ins 5 (offspring 1)

Ins 6 (offspring 2)

Insect 1

Insect 3

Ins 7

Ins 8

2nd Iteration :

Ins 1

Ins 3

Ins 5

Ins 6

Ins 8

Ins 9

Ins 10

Steps of Genetic Algorithm

- i) Set up an initial population
- ii) Check fitness of the chromosomes
- iii) crossover (generating new population from initial population)
- * iv) Mutation
- check goal

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

N-Queen Problem

initial population

0	1	2	3
4	3	4	1

→ chromosome 1

0	1	2	3
4	3	3	4

→ chromosome 2

0	1	2	3
4	2	1	3

→ chromosome 3

0	1	2	3
2	2	4	4

→ chromosome 4

N=4 Queen Problem

4	Q		Q	
3		Q		
2				
1				Q
0				

Vertically / Horizontally /

Diagonally clash

not valid

4	Q			Q	
3		Q	Q		
2					
1					
0					

8 Queen Problem

1) Genetic Algorithm

i) Generate Initial Population

ii) check fitness of the chromosome

no of attacking pair = 0

iii) Crossover → Generate new chromosome

iv) Mutation

8							
7							
6							
5							
4	Q	Q					Q
3							
2	Q					Q	
1				Q	Q		
0							

TOPIC NAME :

DAY :
TIME :
DATE : / /

Initial population

Chromosome 1:	2	4	1	5
Chromosome 2:	3	9	5	4
Chromosome 3:	1	9	3	2
Chromosome 4:	5	4	2	3

Fitness function: No of non attacking pair (fitness value) $\Rightarrow$ element

No of attacking pair = $3C_2 + 1 + 1$

$$+ 1 + 1 + 1 = 8$$

Total Pair = $8C_2 = 28$

Chromosome 2: 5

$\therefore$ Non attacking pair = $28 - 8 = 20$

Chromosome 2 $\Rightarrow$ 5

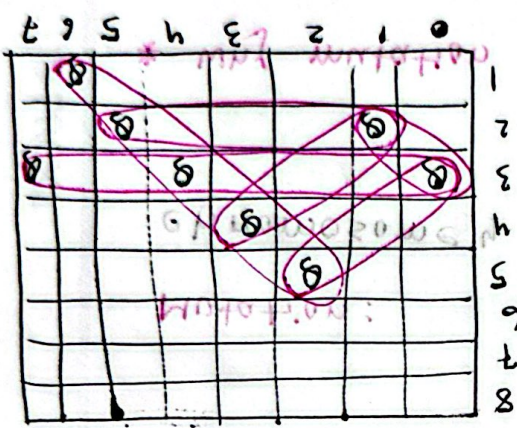
No of attacking pair = $3C_2 + 1$

$+ 1 + 5C_2 + 1 + 1$

$= 17$

Non. Attacking pair = $28 - 17 = 11$

fitness value low 27m 27m 27m



TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Crossover:

Single point crossover:

Chromosome 1: 2 4 4 | 1 5 1 2 4
Chromosome 3: 1 2 2 | 3 5 4 3 7

Chromosome 5: 2 4 4 3 5 4 3 7
Chromosome 6: 1 2 2 1 5 1 2 4

Two point crossover:

Chromosome 3: 1 2 | 2 3 | 5 4 3 7
Chromosome 4: 5 4 | 2 3 | 1 1 3 7

Chromosome 7: 1 2 2 3 5 4 3 7

Chromosome 8: 5 4 2 3 1 1 3 7

Mutation:

Chromosome 4: 5 4 2 4 1 1 3 7

* why mutation is important?

↳ fitness check

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : 05 / 07 / 25

0/1 Knapsack Problem using Genetic Algorithm

Objective: Maximize the value of items packed in knapsack without exceeding weight limit.

Constraint: Max Mass = 10 kg

Item	Mass(Kg)	Cost(₹)
1	6	30
2	3	14
3	4	16
4	2	9

Population:

Chromosome 1: 0101

fitness:

23

Chromosome 2: 1010

46

Chromosome 3: ~~0101~~ 0111

39

Chromosome 4: 1100

44

Crossover:

Chromosome 2: 1010

Chromosome 4: 1100

Mutation

Chromosome 2: 1010

Chromosome 3: 0101

Chromosome 4: 1100

Chromosome 5: 1000

Goal test

TOPIC NAME: _____

DAY: _____

TIME: _____ DATE: / /

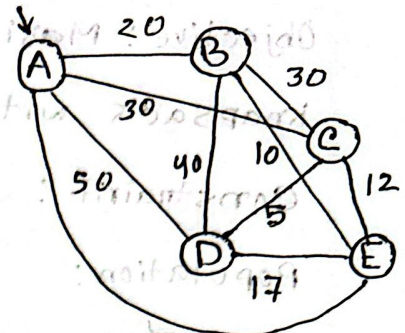
TSP (Travelling Salesman Problem)

start point

travel each states only once

return to the initial state

Shortest distance



Solve this using Genetic Algorithm

Population:

Fitness ($1/\text{distance}$)

Chromosome 1: A B C D E A

Chromosome 2: A B D C E A

Chromosome 3: A C B D E A

Chromosome 4: A D B C E A

Chromosome 1: 1 2 3 4 5 7 6

Chromosome 2: 1 7 2 3 4 6

1 2 3 4 5 7 6 1

1 2 3 4 6 7 5 1

Mutation:

A ~~B~~ C ~~D~~ E A

TOPIC NAME : _____

DAY : _____

TIME : _____

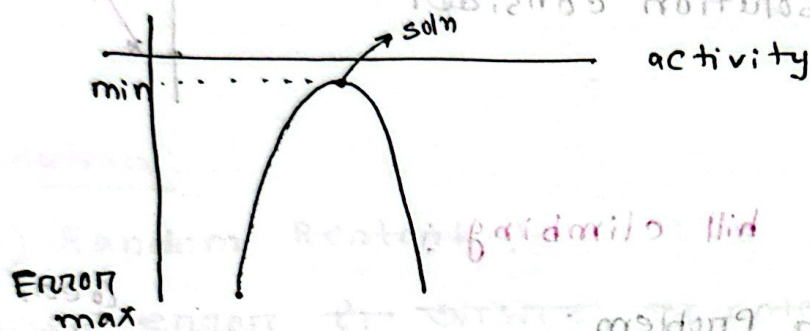
DATE : 12/07 / 25

(local) search: when no exact goal / solution

Infinite searching space

Do not keep track of the path

Hill climbing Algorithm



Advantage of LS:

→ less memory

→ less time

→ Applicable for large size graph

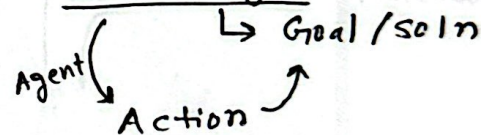
Local search:

↳ No Exact Goal / solution

↳ No need to keep track

↳ Domain infinite

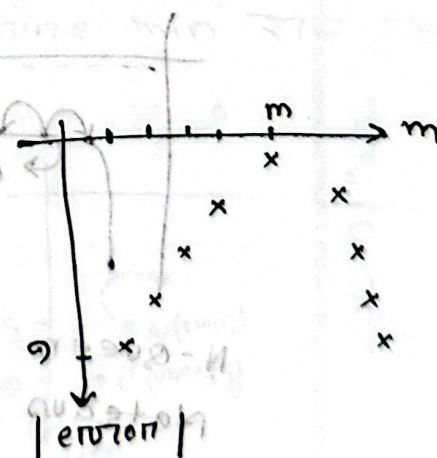
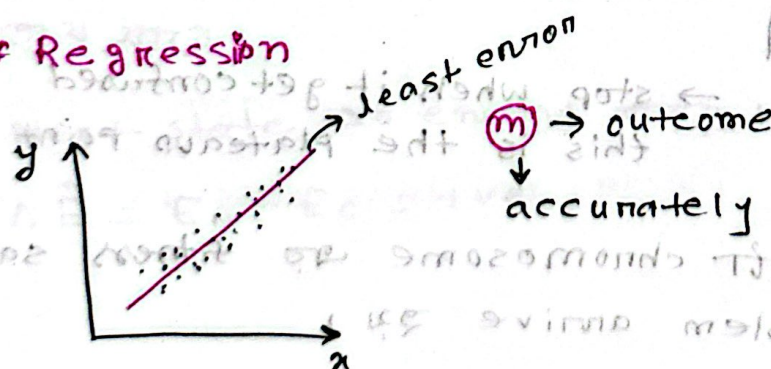
Searching



↳ Nodes trade

Hill climbing

* Regression



Advantage:

i) Less memory requirement

ii) scalable, works properly

TOPIC NAME: _____

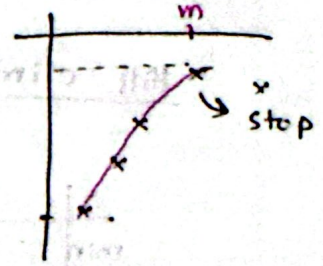
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TIME: _____

DATE: / /

Hill climbing: শুধুমাত্র এখানে কমাড়, শুধুমাত্র consider করতে। কিন্তু যখন এখানে হঠাৎ বাড়ার শুরু হবে, (যদি) মনে হবে এ point এ থাকা যাবে।

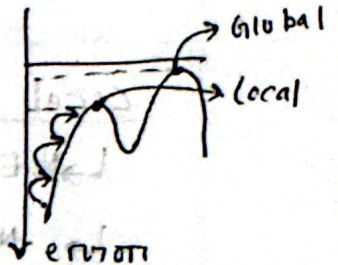
disadvantage: এখানে একবার বাড়ার শুরু হলেই stop হয়ে যায়। next কোনো better solution consider করা না।



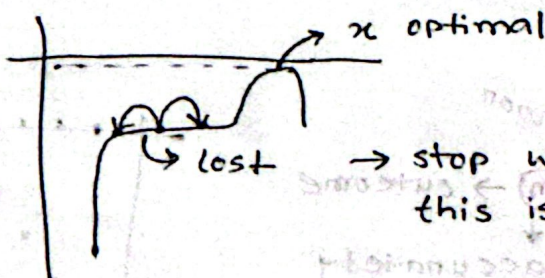
Problem with hill climbing:

i) Local Maxima Problem:

local vs global } maxima



ii) Plateaus:



→ stop when it get confused
this is the plateau point

N-Queen এ ৭টি chromosome এর fitness same হলে plateau problem arrive হয়।

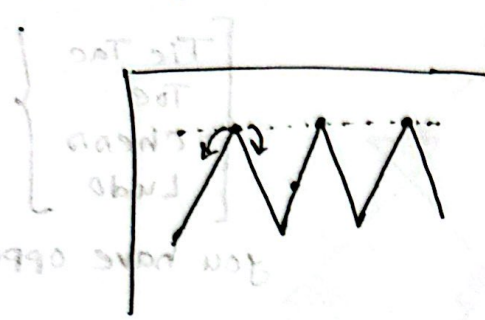
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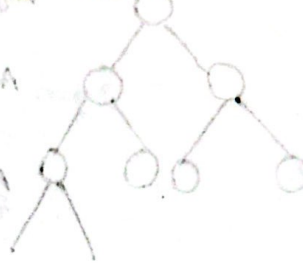
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DATE: / /

iii) Ridge : better than outcome যদি পাই, তাই random



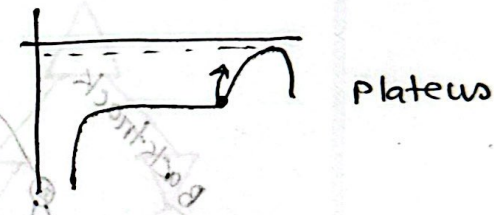
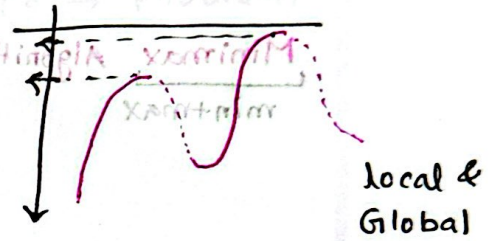
আরেকটা point থেকে শুরু করতে, কিন্তু কিছুদূর পর আবার same issue arrive হবে।



Solution:

i) Random Restart :

যে solution দে আনবে, তা note করে রাখতে। এরপর আবার random point থেকে শুরু করে যদি better soln পাই, মেরি রাখতে।

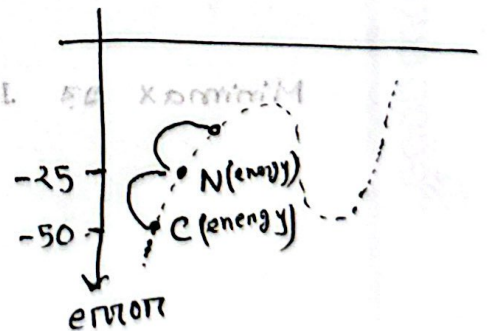


ii) Problem Reformulation : hill এর shape change হয়

iii) Simulated Annealing : Temperature high হলে high error ও consider করে। Temperature min হলে stop হয়ে থাকে।

Next state এর energy বৃদ্ধি।

$$\Delta E = E_N - E_C = +ve$$



TOPIC NAME : _____

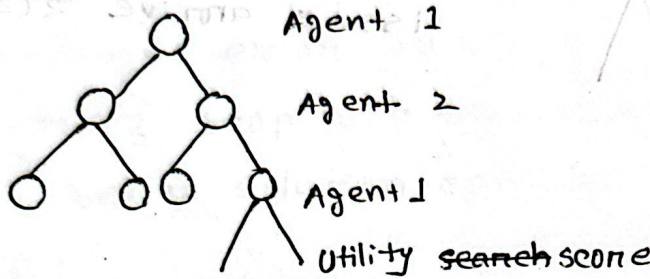
DAY : _____

TIME : _____

DATE : 17 / 7 / 25

Adversarial Search

→ Game playing



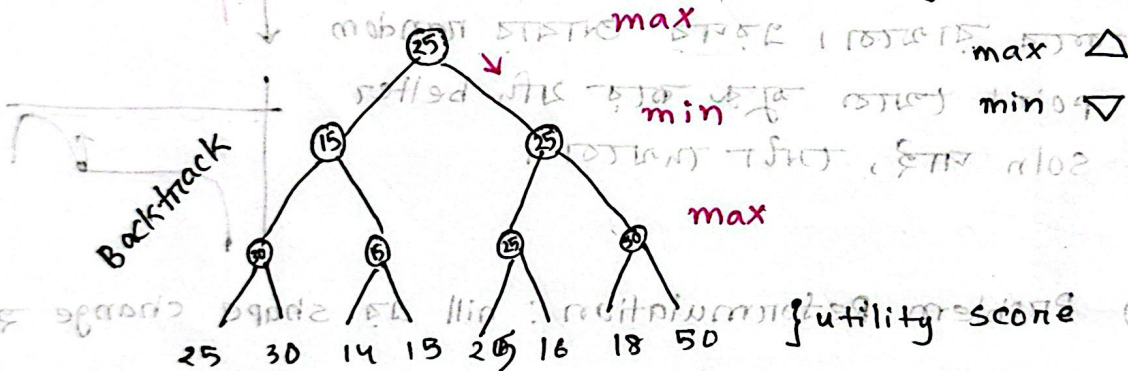
{ Tic Tac
 Toe
 Chess
 Ludo } 2 players
 you have opponent

Minimax Algorithm

→ Lookahead

min+max

Agent 1 vs Agent 2



Agent 1 to consider max agent 1 max
opponent min

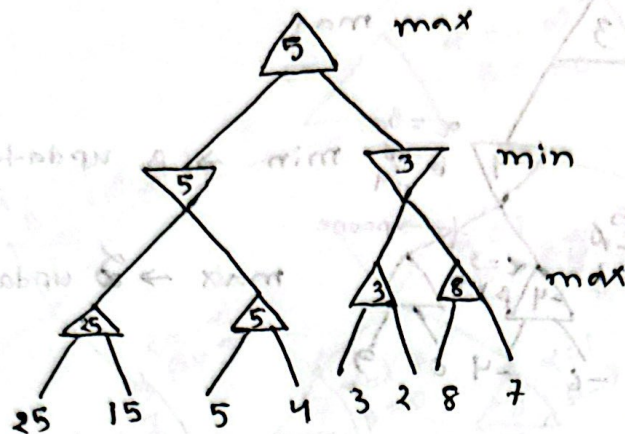
Agent 2 " " " " " agent 2 max
agent 1 min (opponent)

Minimax is 1st player to max consider

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /



Minimax Algorithm সব node explore করা $\rightarrow$ problem
Solution $\rightarrow$ Alpha beta Pruning

Alpha Beta Pruning: computation time save করতে পারি

$\alpha \rightarrow -\infty$

$\beta \rightarrow +\infty$

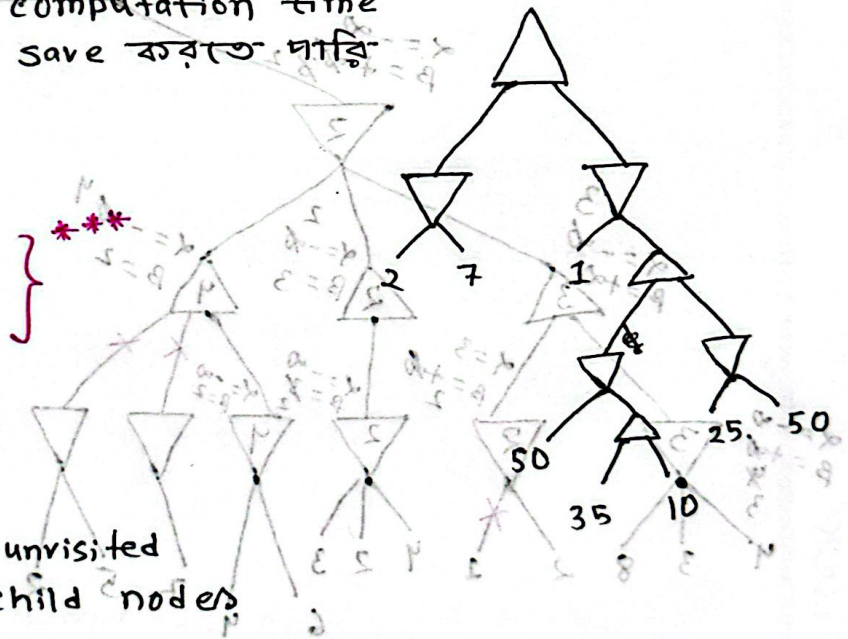
max $\rightarrow$ α update

min $\rightarrow$ β update

Pruning condition :

if $\alpha \geq \beta$:

prune all the unvisited child nodes

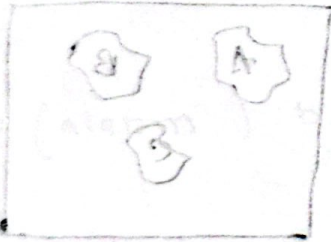
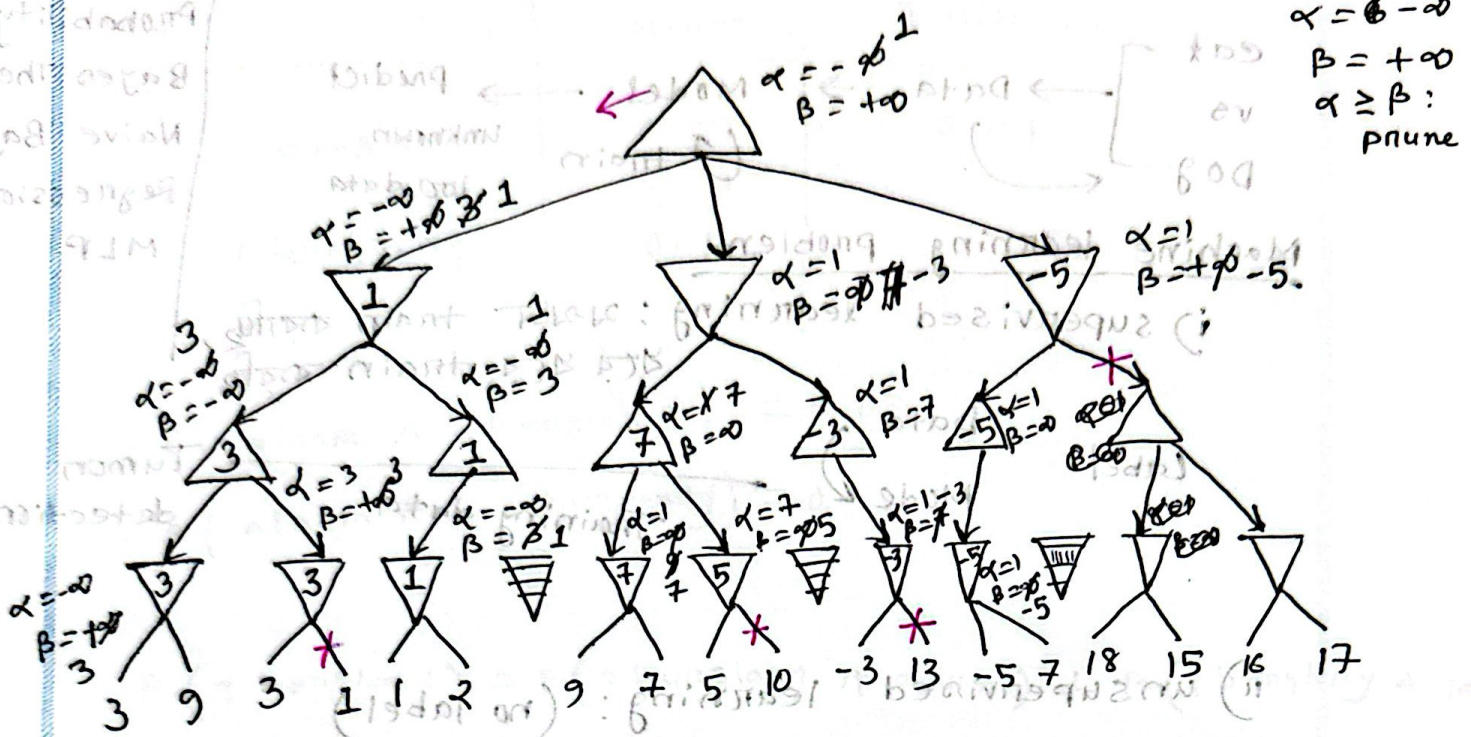


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DAY : _____

TIME : _____

DATE : 19 / 7 / 25



2nd level

Product A
 Product B
 Product C